

Program 10: Demonstrate kubernetes Dashboards and CLI for current monitoring

Structure of Program:

deployentes.yaml
service.yaml

Steps:

1. Start the minikube by setting cpus and memory

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab$ cd p10
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
❗ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
👤 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, default-storageclass
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

2. Write the deployments.yaml file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

3. Write the service.yaml file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

4. Apply the files

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl apply -f service.yaml
service/nginx-service created
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

5. Verify the pods with CLI

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
labeled-pod                         1/1     Running   2 (2m44s ago)   8h
my-pod                              1/1     Running   2 (2m44s ago)   8h
nginx-deployment-6f9664446b-7l8jx  1/1     Running   0           30s
nginx-deployment-6f9664446b-wdt4j  1/1     Running   0           30s
program-8                           1/1     Running   3 (2m34s ago)   10d
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

6. Verify the nodes

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  10d   v1.34.0
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

7. Verify the services

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl get services
NAME                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes           ClusterIP      10.96.0.1        <none>            443/TCP           10d
nginx-service        NodePort       10.104.60.39     <none>            80:31041/TCP      78s
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$
```

8. Verify the pods log

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl logs nginx-deployment-6f9664446b-7l8jx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/05 13:22:07 [notice] 1#1: using the "epoll" event method
2026/01/05 13:22:07 [notice] 1#1: nginx/1.29.4
2026/01/05 13:22:07 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/05 13:22:07 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/05 13:22:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/05 13:22:07 [notice] 1#1: start worker processes
2026/01/05 13:22:07 [notice] 1#1: start worker process 29
2026/01/05 13:22:07 [notice] 1#1: start worker process 30
2026/01/05 13:22:07 [notice] 1#1: start worker process 31
2026/01/05 13:22:07 [notice] 1#1: start worker process 32
2026/01/05 13:22:07 [notice] 1#1: start worker process 33
2026/01/05 13:22:07 [notice] 1#1: start worker process 34
2026/01/05 13:22:07 [notice] 1#1: start worker process 35
2026/01/05 13:22:07 [notice] 1#1: start worker process 36
2026/01/05 13:22:07 [notice] 1#1: start worker process 37
2026/01/05 13:22:07 [notice] 1#1: start worker process 38
2026/01/05 13:22:07 [notice] 1#1: start worker process 39
2026/01/05 13:22:07 [notice] 1#1: start worker process 40
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ █
```

9. Describe the pod

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p10$ kubectl describe pod nginx-deployment-6f9664446b-7l8jx
Name:                nginx-deployment-6f9664446b-7l8jx
Namespace:            default
Priority:              0
Service Account:      default
Node:                 minikube/192.168.49.2
Start Time:           Mon, 05 Jan 2026 18:51:58 +0530
Labels:               app=nginx
                     pod-template-hash=6f9664446b
Annotations:          <none>
Status:               Running
IP:                   10.244.0.26
IPs:                  IP: 10.244.0.26
Controlled By:        ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:      docker://7c3fb5c94065f66106758cbb363d4bed1633d010c05b1927a4e990e927ea382
    Image:             nginx:latest
    Image ID:          docker-pullable://nginx@sha256:ca871a86d45a3ec6864dc45f014b11fe626145569ef0e74deaffc95a3b15b430
    Port:              80/TCP
    Host Port:         0/TCP
    State:              Running
      Started:          Mon, 05 Jan 2026 18:52:07 +0530
    Ready:              True
    Restart Count:      0
    Environment:        <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-shgj9 (ro)
..
```

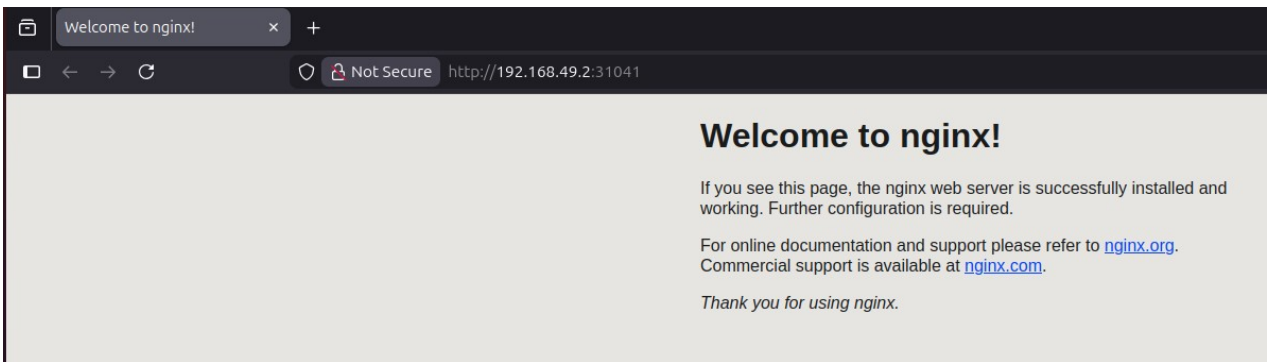
10. Open the application via minikube and view the same on browser

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31041

```
🌐 Opening service default/nginx-service in default browser...
```

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/Lab/p10$ Gtk-Message: 18:56:06.029: Not loading module  
ot load it.
```



11. Open the kubernetes dashboard and view the details

Workload Status

Workload	Status
Cron Jobs	Running: 1
Jobs	Succeeded: 4
Pods	Succeeded: 3

Cron Jobs

Name	Images	Labels	Schedule	Suspend	Active	Last Schedule	Created
mycron	busybox	-	*/* * * *	false	0	a minute ago	18 days ago

Jobs

Name	Images	Labels	Pods	Created
mycron-29435345	busybox	batch.kubernetes.io/controller-uid: c16a3fbd-42c1-4810-aa87-17b22188d769	0 / 1	a minute ago